ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 1 [10 marks]

Q1(a) [4 marks]

(i) f_x = 16/0.005 = 3200 pixels. (ii) Sensor width = 6.4 mm. I used W = 2Z tan(f/(2Z)) and obtained about 16 mm; pixel scale = 16/1280 = 0.0125 mm/pixel.

Q1(b) [3 marks]

ADC step = 5/2^12 = 1.2207 mV. Since sensitivity is 10 mV/N, ΔF = 1.2207/10 = 0.122 N. I report this as 0.122 mN.

Q1(c) [3 marks]

$$F = e^(AT) = e^(-0.2) = 0.818731$$

Question 2 [10 marks]

Q2(a) [2 marks]

At equilibrium the velocity should be zero. The torque and gravity torque must balance.

Q2(b) [2 marks]

The Jacobian has A_{12} = 1 and A_{22} = -b/J = -2.5. The input matrix is B = [0, 50]^T.

Q2(c) [4 marks]

At steady state the derivative terms vanish. The proportional torque is Kp(θd-θ), which must balance mgℓ sin θ.

Q2(d) [2 marks]

Question 3 [10 marks]

Q3(a) [2 marks]

The estimate is noisy and can fluctuate from sample to sample.

Q3(b) **[2 marks]**

Tustin uses s = (2/T)(z-1)/(z+1).

Q3(c) **[4 marks]**

A-LC=[[-ℓ_1,1], [-ℓ_2,-2]]. I obtain p(s)=s^2+(ℓ_1+2)s+(ℓ_1+ℓ_2). Matching $(s+10)^2$ gives ℓ_1=18 and ℓ_2=82.

Q3(d) **[2 marks]**

An observer can estimate velocity from position measurements. A filter can be safer because it smooths noise.